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ADRIAIndicators.jl: a Julia package for summarizing reef ecological model outputs

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Software

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Summary

ADRIAIndicators.jl is a Julia package for analyzing outputs from coral reef ecological models. Its primary purpose is to provide a standardized and dependency-free toolkit for transforming high-dimensional model outputs such as coral abundance by reef, time, species, and size class into lower-dimensional, interpretable metrics. The package offers a wide range of functions, from simple aggregations and unit conversions to more complex indices and estimators derived from regression models. These tools help with the estimation of functional diversity, juvenile abundance, shelter volume, fish biomass, and overall reef condition, enabling consistent and comparable analysis across different coral ecology models such as CoralBlox (Ribeiro de Almeida et al., 2024), C~Scape (Cresswell et al., 2024), ReefMod (Bozec et al., 2022), and CoCoNet (Condie, 2022).

Statement of Need

Models of coral reef ecosystems often produce large volumes of high-dimensional data. There is a need for standardized tools to summarize and analyze these model outputs to facilitate inter-model comparison of environmental projections and communicate results to managers and stakeholders. ADRIAIndicators.jl provides a set of standard indicator metrics that can be used to summarize reef state in ecological model outputs. These were originally implemented within the ADRIA.jl Decision Support package (Iwanaga et al., 2025) but were being reproduced in many workflows that did not use ADRIA.jl. Separating these indicators from ADRIA.jl is part of efforts to better compartmentalize code in a more reusable and interoperable manner.

ADRIAIndicators.jl is written in Julia (Bezanson et al., 2017), a high-level, high-performance programming language for technical computing. This package is designed to be easy to use, and provides an in-place option for all metrics for any eventual wrappers that may be implemented in other languages such as Python and R. Such wrappers could be developed leveraging Julia's support for language interoperability and compilation capabilities (such as those provided by JuliaC.jl).

Available Indicators

The indicators implemented in ADRIAIndicators.jl are classified into three categories: Aggregations, Conversions, and Metrics. Aggregations are convenience methods for reducing the

dimensionality of data by summarizing arrays. Conversions handle transformations between different units or representations of coral cover. Metrics derive higher-level, interpretable indicators from the raw model data, such as coral diversity, shelter volume, and composite indices for reef health.

Metric Name	Type	Reference
Relative Cover	Aggregation	
Relative Location Taxonomy Cover	Aggregation	
Relative Taxonomy Cover	Aggregation	
LTMP Cover	Aggregation	
LTMP Location Taxonomy Cover	Aggregation	
LTMP Taxonomy Cover	Aggregation	
Relative Juveniles	Aggregation	
Relative Location Taxonomy Juveniles	Aggregation	
Relative Taxonomy Juveniles	Aggregation	
Relative Habitable Cover to Reef Cover	Conversion	
Reef Cover to Relative Habitable Cover	Conversion	
Absolute Shelter Volume	Metric	(Aston et al., 2022; Urbina-Barreto et al., 2021)
Relative Shelter Volume	Metric	-
Coral Diversity	Metric	(Hill, 1973)
Coral Evenness	Metric	-
Reef Condition Index	Metric	(Heneghan et al., 2025)
Reef Fish Index	Metric	(Graham & Nash, 2013)

A dash (-) in the 'Reference' column indicates the reference is the same as the entry directly above it.

Usage

The order of dimensions is always the same in ADRIAIndicators.jl,

1. Time
2. Groups
3. Sizes
4. Locations
5. Scenarios

If a dimension is missing then the order remains the same however the missing dimensions are excluded. Furthermore, all metrics have an option to provide a buffer as input in the cases where one wants to write the metric into an existing array or sub-array. This implementation is relied upon by functions that allocate the returned array as-well and was chosen to account for any any decisions in the future where another language may wrap this library and need to pass memory that is not managed by Julia.

```julia
using ADRIAIndicators

# Create some dummy model output data
n_timesteps = 75
n_groups = 5
n_sizes = 7
```

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```julia
n_locations = 3806

# Raw model coral cover outputs with dimensions [timesteps · groups · sizes · locations]
raw_model_cover = rand(Float64, n_timesteps, n_groups, n_sizes, n_locations);

# Juveniles mask with dimensions [sizes]
is_juvenile = [true, true, false, false, false, false, false];

# Calculate and allocate new array for metric
rel_juveniles = relative_juveniles(raw_model_cover, is_juvenile);

# Users can provide output buffers if it is more convenient for them
rel_juveniles_out = zeros(Float64, n_timesteps, n_locations);
relative_juveniles!(raw_model_cover, is_juvenile, rel_juveniles_out);
```


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